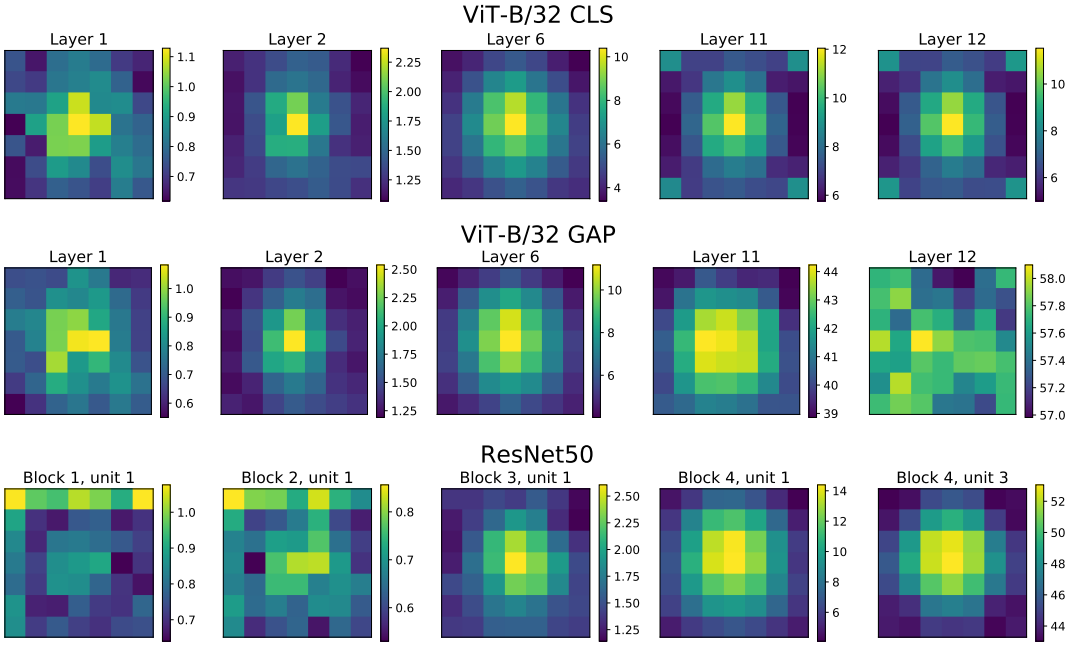


Figure D.4: Linear probes spatial localization. We train a linear probe on each individual token and plot the average accuracy over the test set, in percent. Here we plot the results for each token a subset of layers in 3 models: ViT-B/32 trained with a classification token (CLS) or global average pooling (GAP), as well as a ResNet50. Note the different scales of values in different sub-plots.

E Additional Representation Propagation Results

Figure E.1 shows the ratio of representation norms between skip connections and MLP and Self-Attention Blocks. In both cases, we observe that the CLS token representation is mostly unchanged in the first few layers, while later layers change it rapidly, mostly via MLP blocks. The reverse is true for the spatial tokens representing image patches, whose representation is mostly changed in earlier layers and does not change much during later layers. Looking at the cosine similarity of representations between output in Figure E.2 confirms these findings: while spatial token representations change more in early layers, the output of later blocks is very similar to the representation present on the skip connections, while the inverse is true for the CLS token.

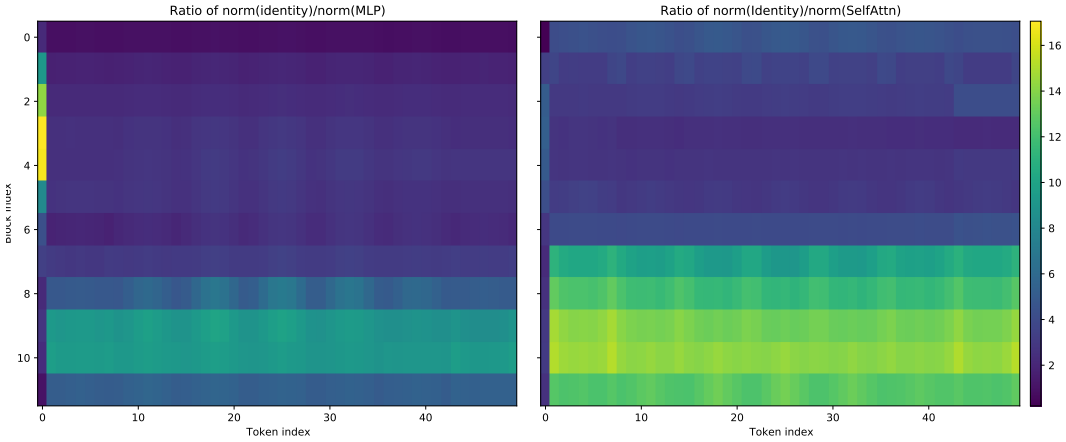


Figure E.1: Additional Heatmaps of Representation Norms Ratio Representation Norms $||z_i||/||f(z_i)||$ between skip connection and the MLP or Self-Attention block for the hidden representation of each block on ViT-B/32, separately for each Token (Token 0 is CLS).

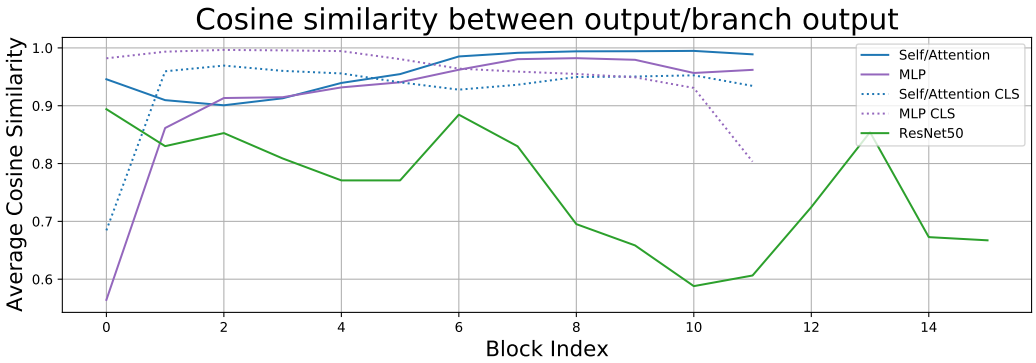


Figure E.2: Most information in ViT passes through Identity Connections. Cosine Similarity of representations between the skip-connection (identity) and the longer branch for ViT-B/16 trained on ImageNet and a ResNet v1. For ViT, we show the CLS token separately from the rest of the representation.

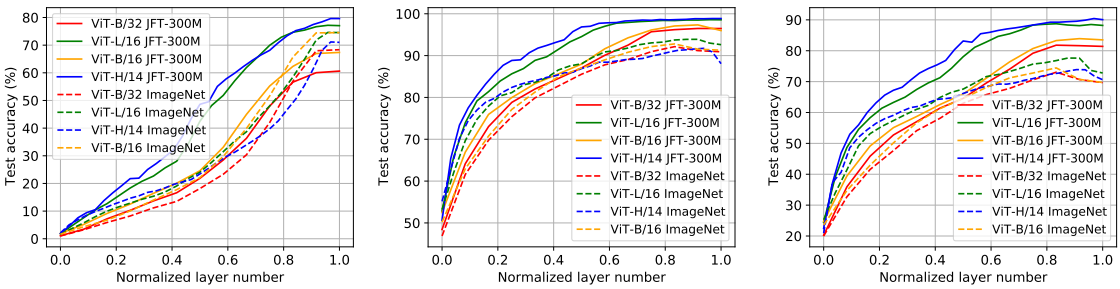
F Additional results on linear probes

Here we provide additional results on linear probes complementing Figures 11 and 13 of the main paper. In particular, we repeat the linear probes on the CIFAR-10 and CIFAR-100 datasets and, in some cases, add more models to comparisons. For CIFAR-10 and CIFAR-100, we use the first 45000 images of the training set for training and the last 5000 images from the training set for validation. Additional results are shown in Figures F.1, F.2, F.3.

Moreover, we discuss the results in the main paper in more detail. In Figure 11 (left) we experiment with different ways of evaluating a ViT-B/32 model. We vary two aspects: 1) the classifier with which the model was trained, classification token (CLS) or global average pooling (GAP), 2) The way the representation is aggregated: by just taking the first token (which for the CLS models is the CLS token), averaging all tokens, or averaging all tokens except for the first one.

There are three interesting observations to be made. First, CLS and GAP models evaluated with their "native" representation aggregation approach – first token for CLS and GAP for GAP – perform very similarly. Second, the CLS model evaluated with the pooled representation performs on par with the first token evaluation up to last several layers, at which point the performance plateaus. This suggests that the CLS token is crucially contributing to information aggregation in the latter layers. Third, linear probes trained on the first token of a model trained with a GAP classifier perform very poorly for the earlier layers, but substantially improve in the latter layers and almost match the performance of the standard GAP evaluation in the last layer. This suggests all tokens are largely interchangeable in the latter layers of the GAP model.

To better understand the information contained in individual tokens, we trained linear probes on all individual tokens of three models: ViT-B/32 trained with CLS or GAP, as well as ResNet50. Figure 11 (right) plots average performance of these per-token classifiers. There are two main observations to be made. First, in the ViT-CLS model probes trained on individual tokens perform very poorly, confirming that the CLS token plays a crucial role in aggregating global class-relevant information. Second, in ResNet the probes perform poorly in the early layers, but get much better towards the end of the model. This behavior is qualitatively similar to the ViT-GAP model, which is perhaps to be expected, since ResNet is also trained with a GAP classifier.



(a) ImageNet (b) CIFAR-10 (c) CIFAR-100

Figure F.1: Experiments with linear probes. Additional results on models pre-trained on JFT-300M and ImageNet (Fig. 13 left) – with the addition of ViT-B models and CIFAR-10/100 datasets.